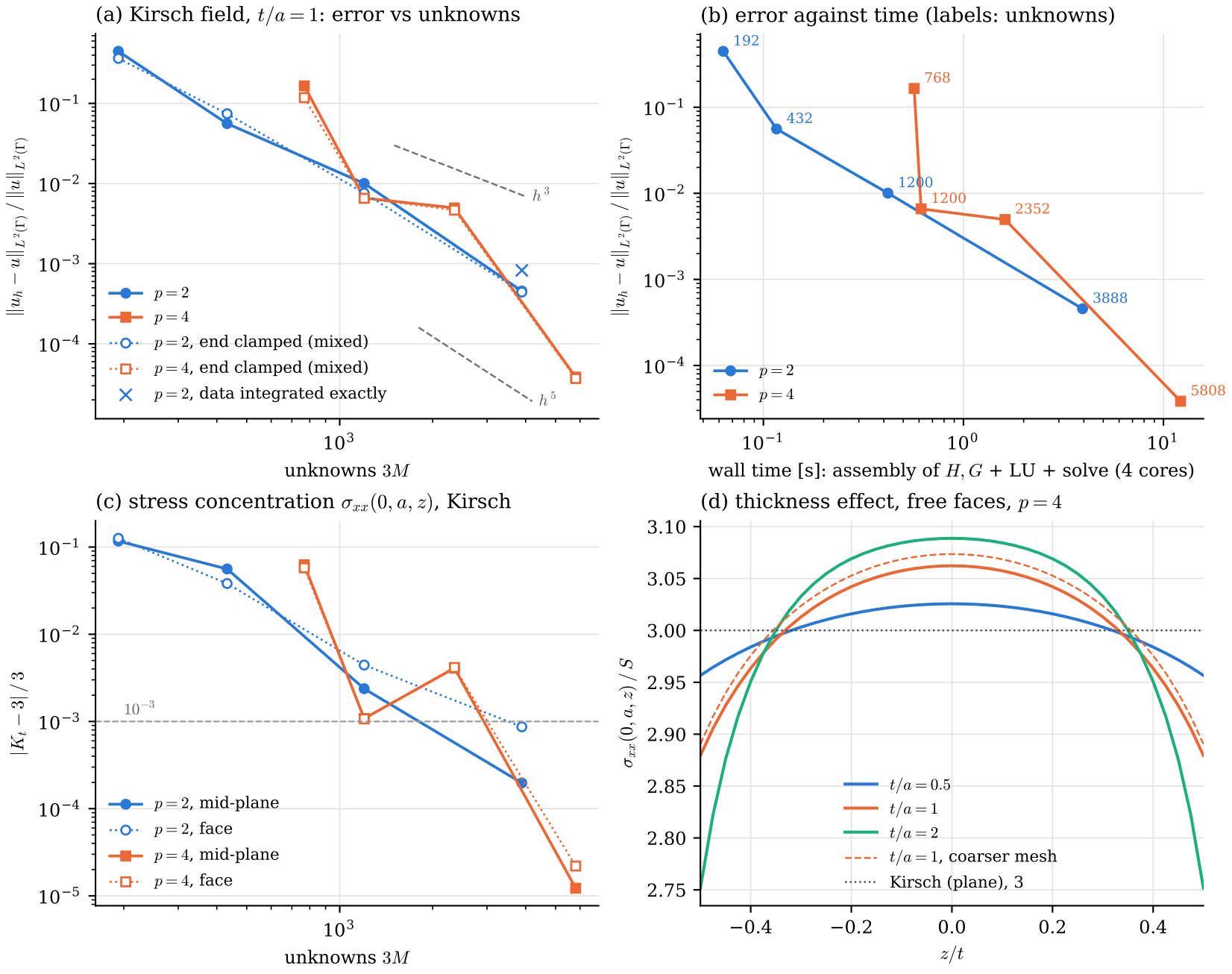
(a) Kirsch field, $t/a = 1$: error vs unknowns
$\|u_h - u\|_{L^2(\Gamma)} / \|u\|_{L^2(\Gamma)}$
unknowns $3M$
h^3
h^5
$p = 2$
$p = 4$
$p = 2$, end clamped (mixed)
$p = 4$, end clamped (mixed)
$p = 2$, data integrated exactly
(b) error against time (labels: unknowns)
192
432
1200
768
1200
2352
3888
5808
wall time [s]: assembly of H, G + LU + solve (4 cores)
(c) stress concentration $\sigma_{xx}(0, a, z)$, Kirsch
$|K_I - 3| / 3$
10^{-3}
$p = 2$, mid-plane
$p = 2$, face
$p = 4$, mid-plane
$p = 4$, face
(d) thickness effect, free faces, $p = 4$
$\sigma_{xx}(0, a, z) / S$
z/t
$t/a = 0.5$
$t/a = 1$
$t/a = 2$
$t/a = 1$, coarser mesh
Kirsch (plane), 3